Fundamental Calculus/Differential and Integral Calculus

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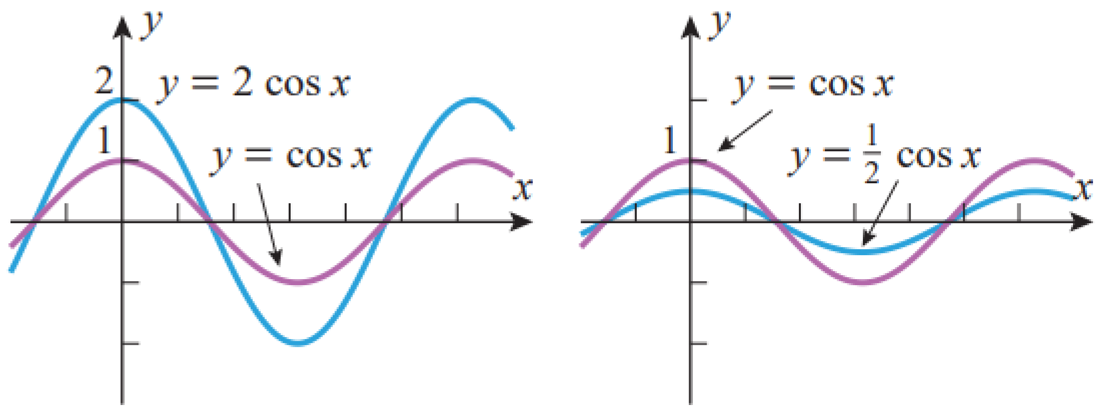
Reference : Calculus 10-th Edition by Howard Anton, Irl Bivens and Stephen Davis

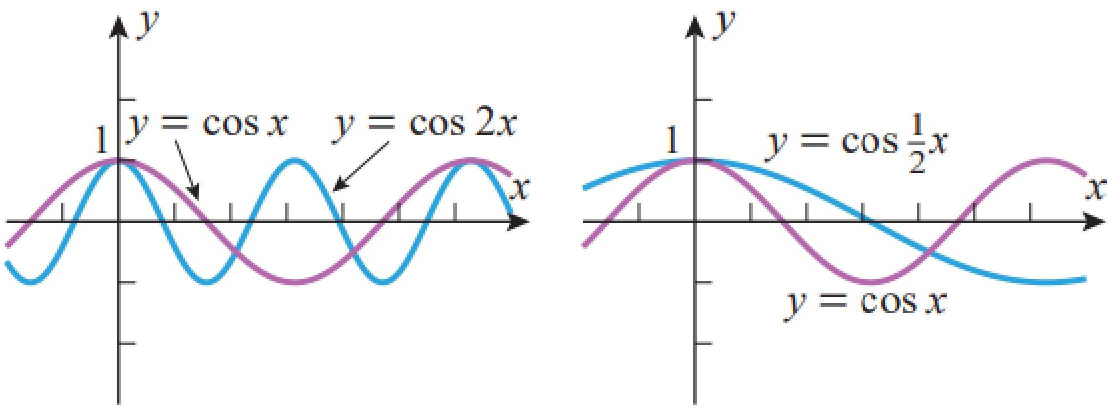
Chapter: 0.2

- New Functions From Old function

STRETCHES AND COMPRESSIONS



STRETCHES AND COMPRESSIONS



Sketch the following two functions. Don't use the graphing calculator or online tool.

(a) $y = 1 + \sin x$

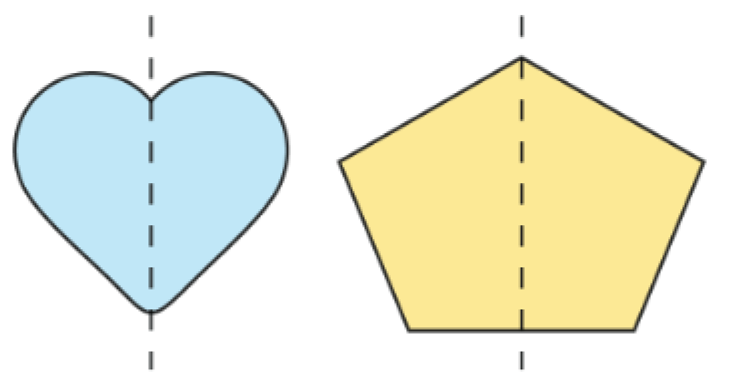
(b) $y = -1 + \cos x$

(c) $y = 1 + \sin \frac{1}{2}x$

(d) $y = 1 + \cos 2x$

Symmetry

- Symmetry is defined as **a balanced and proportionate similarity that is found in two halves of an object**. It means one-half is the mirror image of the other half.

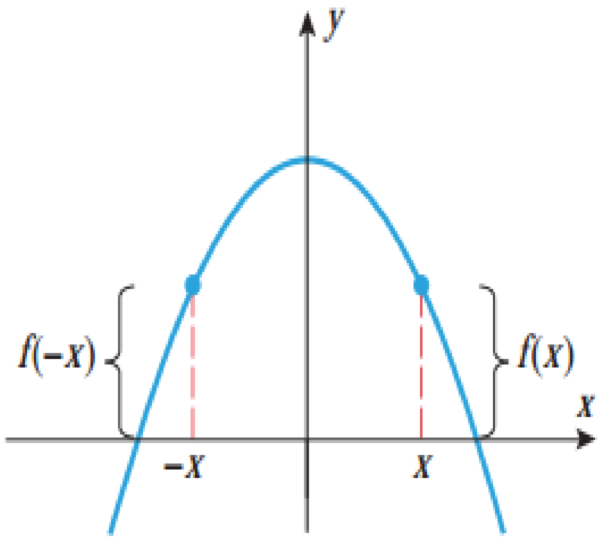


EVEN AND ODD FUNCTIONS

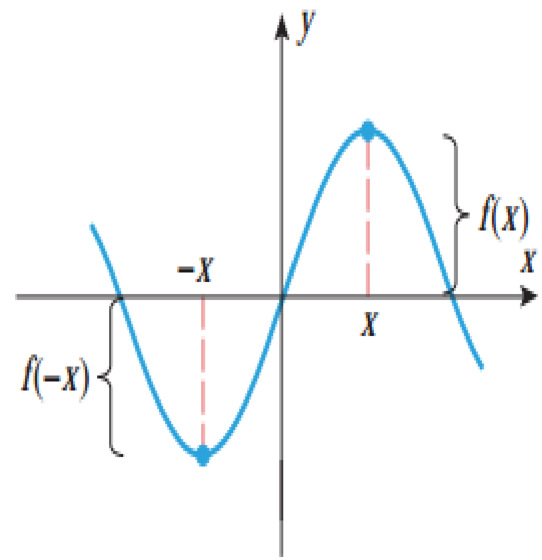
- A function *f* is said to be an *even function* if $f(-x) = f(x)$ and is said to be an *odd function* if $f(-x) = -f(x)$.

- Geometrically, the graphs of even functions are symmetric about the *y*-axis because replacing *x* by $-x$ in the equation $y = f(x)$ yields $y = f(-x)$, which is equivalent to the original.
- Geometrically, the graphs of odd functions are symmetric about the origin.
- Some examples of even functions $are\ x^2, x^4, x^6 and\ cosx$; and some examples of odd functions are$x^3, x^5, x^7, and\ sinx$.

EVEN AND ODD FUNCTIONS



▲ **Figure 0.2.9** This is the graph of an even function since $f(-x) = f(x)$.



▲ **Figure 0.2.10** This is the graph of an odd function since $f(-x) = -f(x)$.

EVEN AND ODD FUNCTIONS

- Determine whether the given function is even or odd?

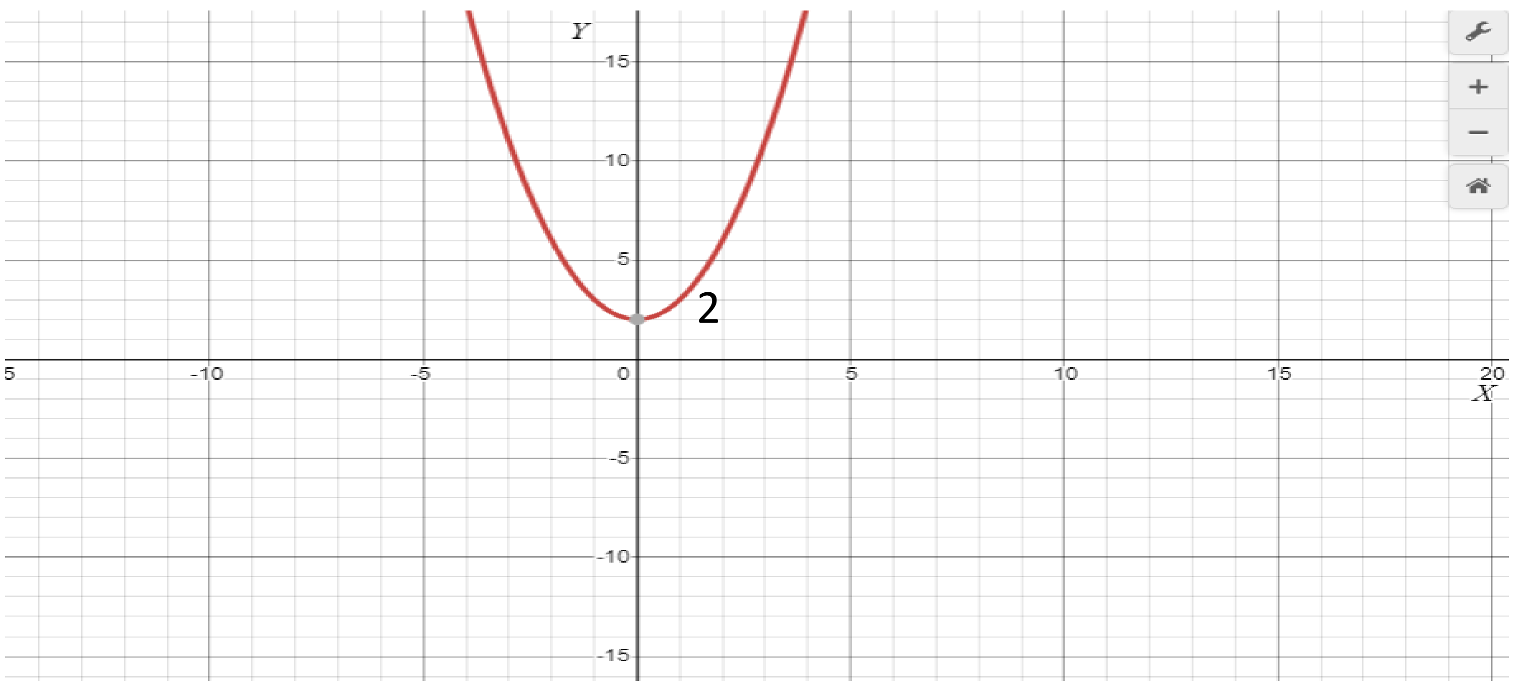
1. $f(x) = x^2 + 2;$
2. $f(x) = x^3 + 2;$
3. $f(x) = \frac{1}{x};$
4. $f(x) = |x| + 2$
5. $f(x) = |x - 1| + 2;$

EVEN AND ODD FUNCTIONS

- Solution: (1)$f(x) = x^2 + 2$;
- Technique:1
- Since, $f(-x) = (-x)^2 + 2$
  $$= x^2 + 2$$
  $$= f(x)$$
- So $f(x)$ is even function.

$y = x^2 + 2$;

- Or, <u>Technique:2</u>
- The graph is symmetric about Y axis, so this function is even function.



EVEN AND ODD FUNCTIONS

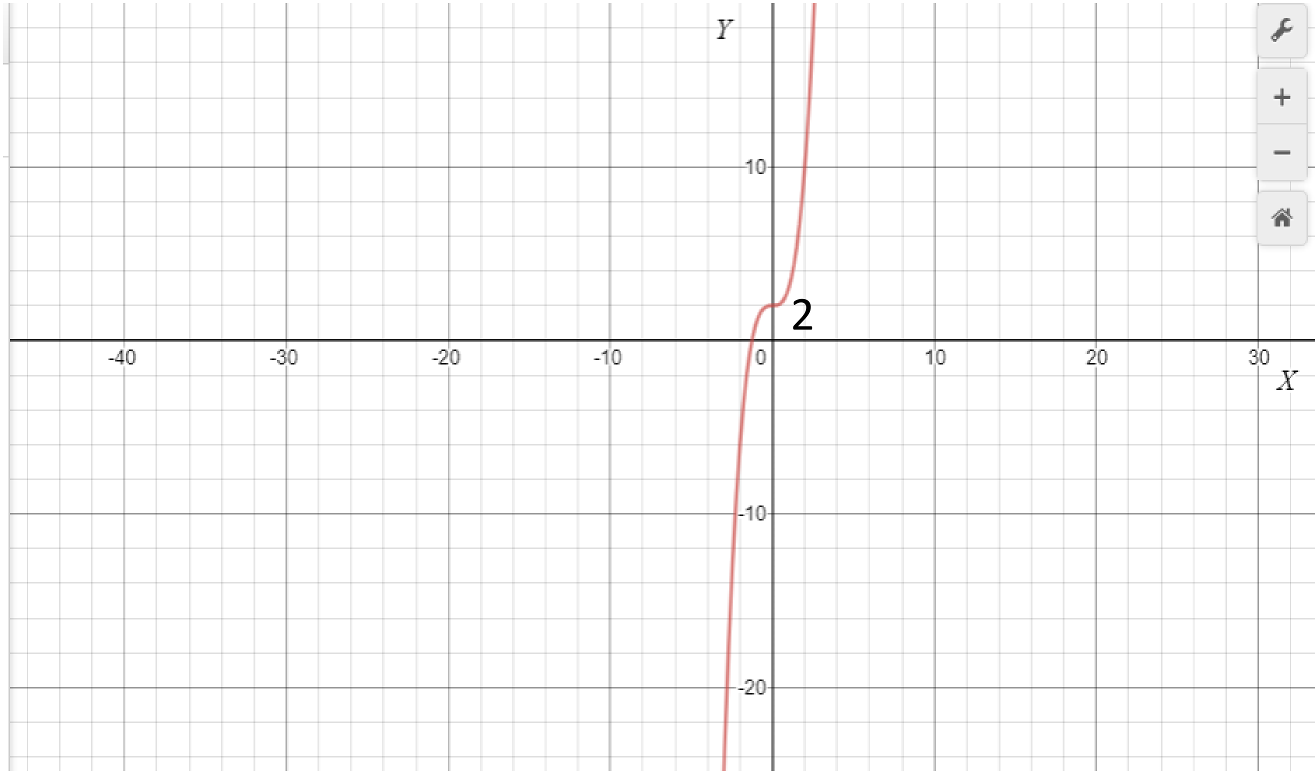
- <u>Solution: (2)</u>
- <u>Technique:1</u>
- Given, $f(x) = x^3 + 2$;
- Since, $f(-x) = (-x)^3 + 2 = -x^3 + 2 \neq f(x) \neq -f(x)$

So $f(x)$ is neither even nor odd function.

EVEN AND ODD FUNCTIONS

Or, <u>Technique:2</u>

The graph is not symmetric about the *y*-axis and also not symmetric about origin, So $f(x)$ is neither even nor odd function.



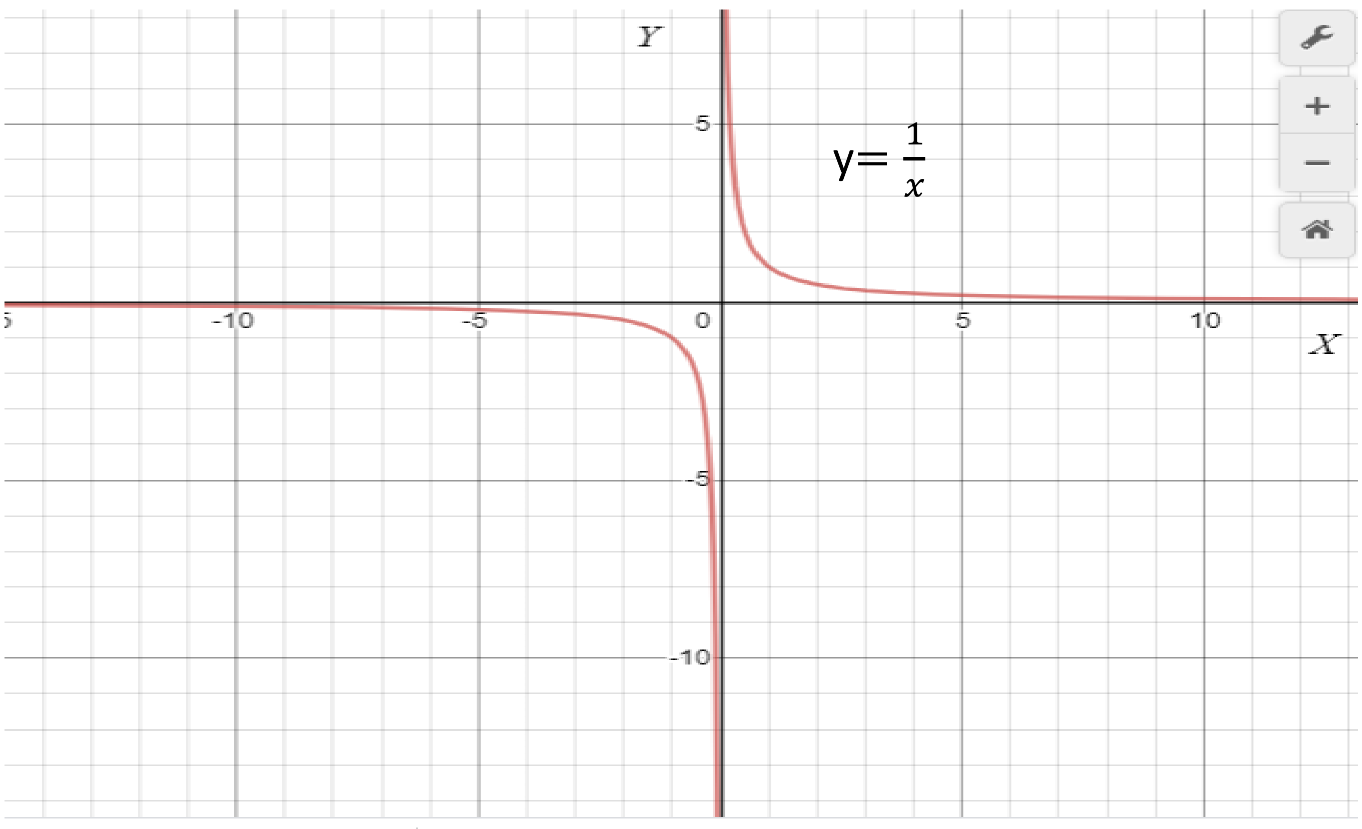
EVEN AND ODD FUNCTIONS

- Solution: (3)
- Technique:1
- Given, $f(x) = \frac{1}{x}$;
- Since, $f(-x) = -\frac{1}{x} = -f(x)$

So $f(x)$ is odd function.

EVEN AND ODD FUNCTIONS

- The graph is symmetric about origin, so this function is odd function.



EVEN AND ODD FUNCTIONS

- Solution: (4)
- Technique:1:
- $f(x) = |x| + 2$;
- Since, $f(-x) = |-x| + 2$

$= |x| + 2 = f(x)$

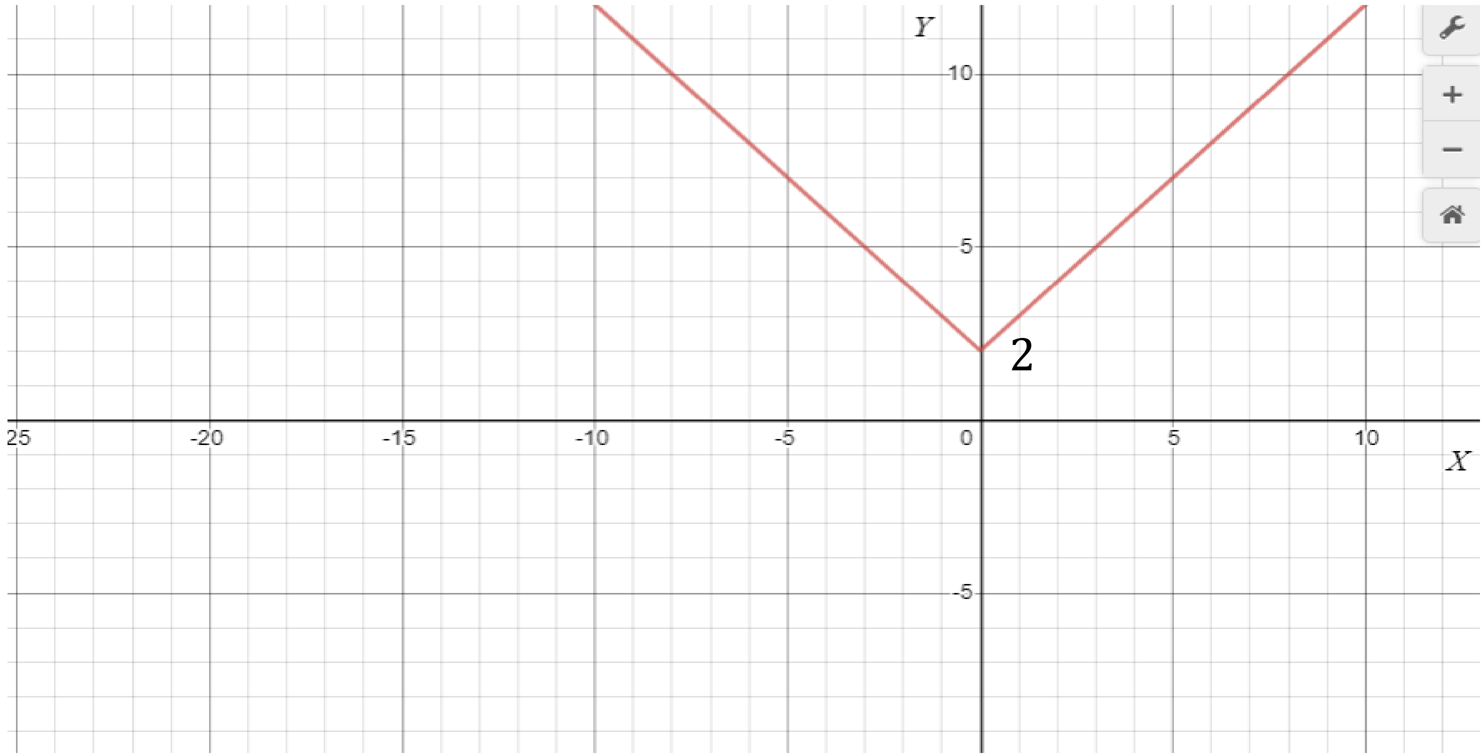
So $f(x)$ is even function.

EVEN AND ODD FUNCTIONS

- Or,

<u>Technique:2</u>

The graph is symmetric about Y axis, so this function is even function.



EVEN AND ODD FUNCTIONS

- Solution: (5)
- Technique:1:
- $f(x) = |x - 1| + 2;$
- Since, $f(-x) = |-x - 1| + 2$

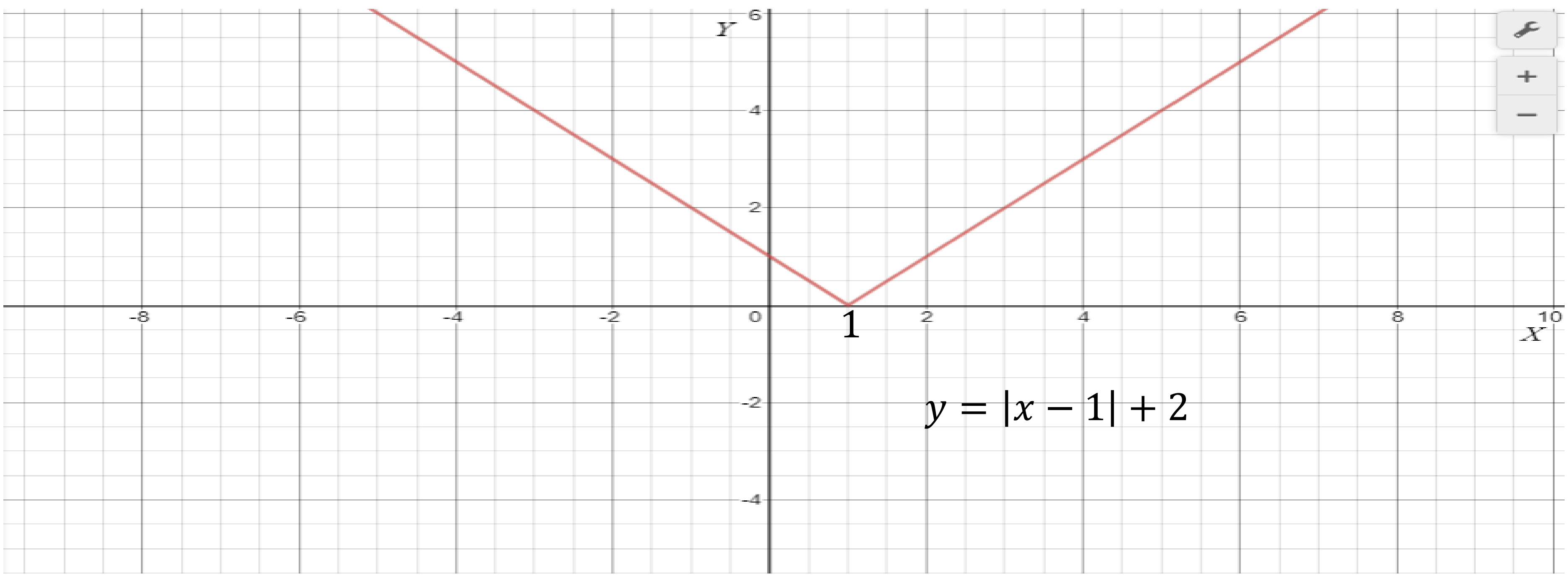
$\neq f(x) \neq f(-x)$

So $f(x)$ is neither even nor odd function.

EVEN AND ODD FUNCTIONS

Or, <u>Technique:2</u>

The graph is not symmetric about the *y*-axis and also not symmetric about origin, So $f(x)$ is neither even nor odd function.



EVEN AND ODD FUNCTIONS

63. In each part, classify the function as even, odd, or neither.

(a) $f(x) = x^2$ (b) $f(x) = x^3$

(c) $f(x) = |x|$ (d) $f(x) = x + 1$

(e) $f(x) = \dfrac{x^5 - x}{1 + x^2}$ (f) $f(x) = 2$

5. Determine whether f is even, odd, or neither. You may wish to use a graphing calculator or computer to check your answer visually. Then sketch them.

(a) $f(x) = \frac{x}{x^2+1}$

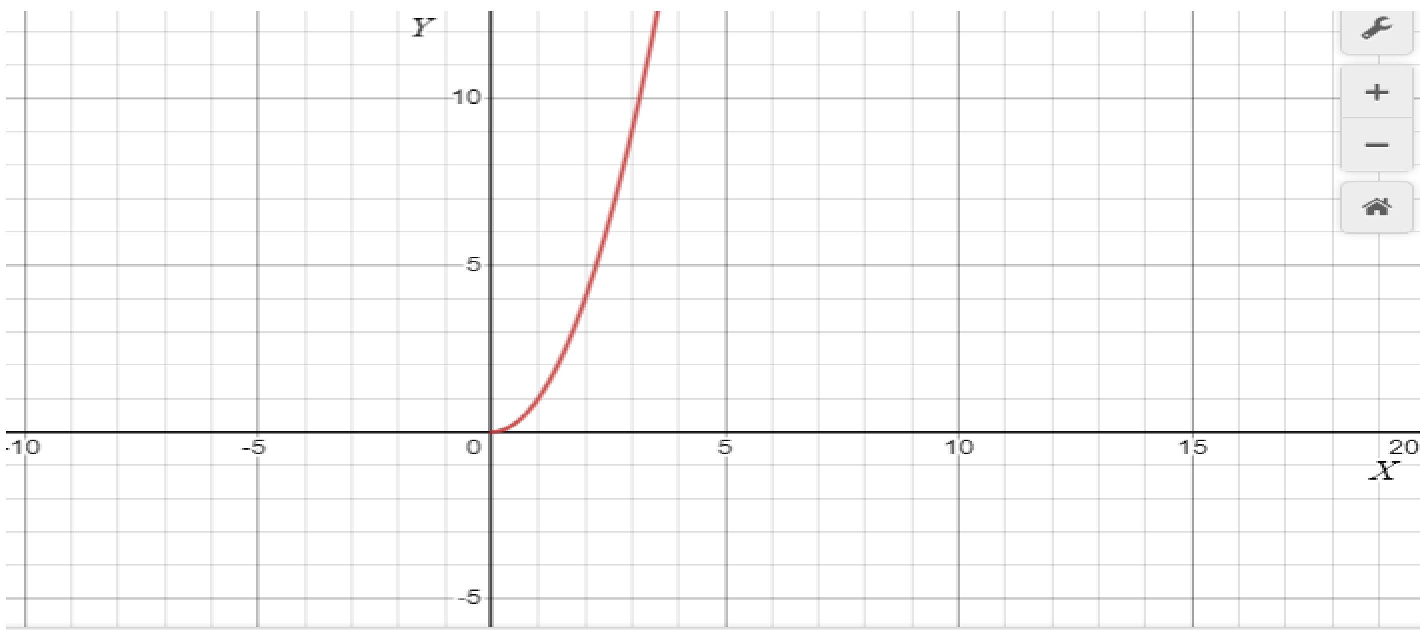
(b) $f(x) = \frac{x^2}{x^4+1}$

(c) $f(x) = x|x|$

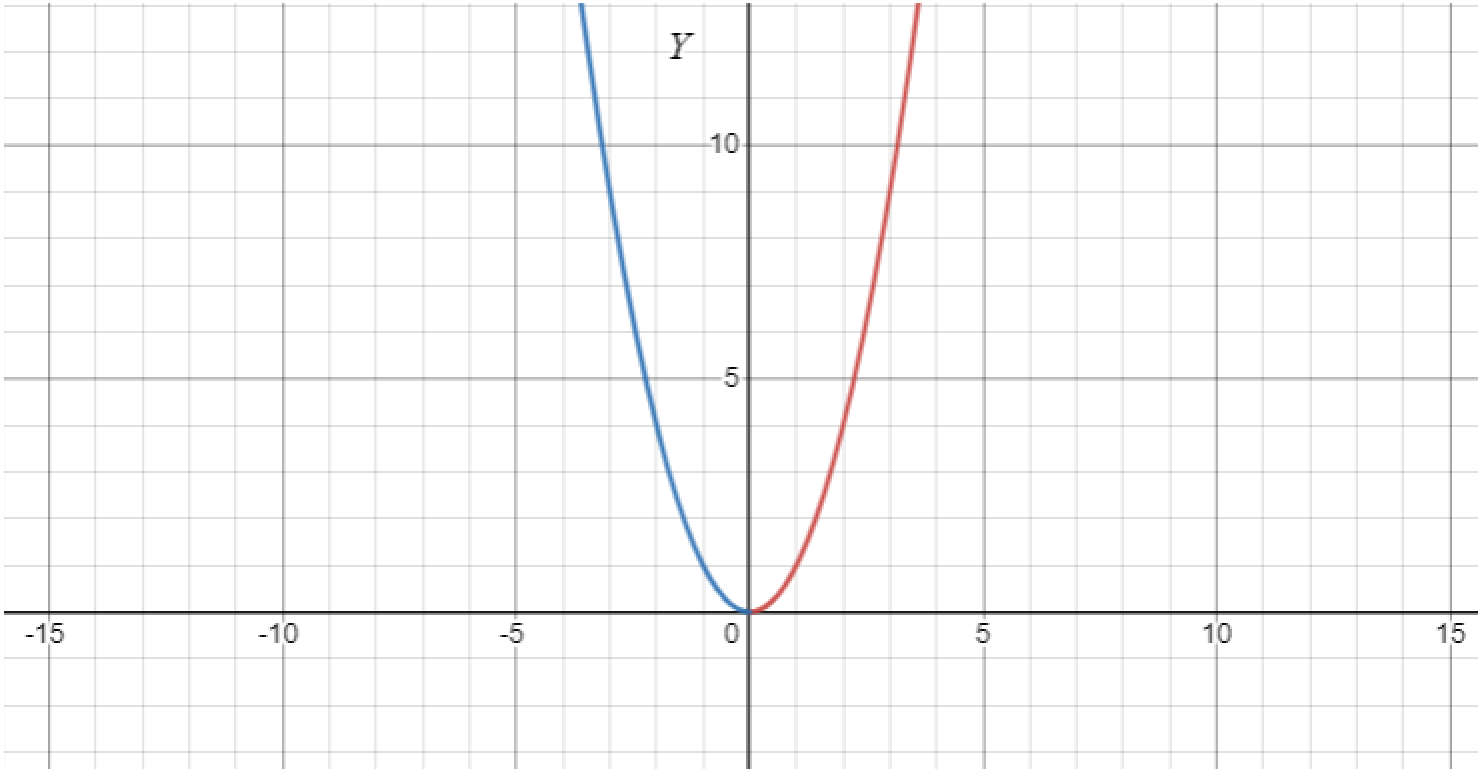
Even and Odd

- Consider $y = x^2; x \geq 0$.

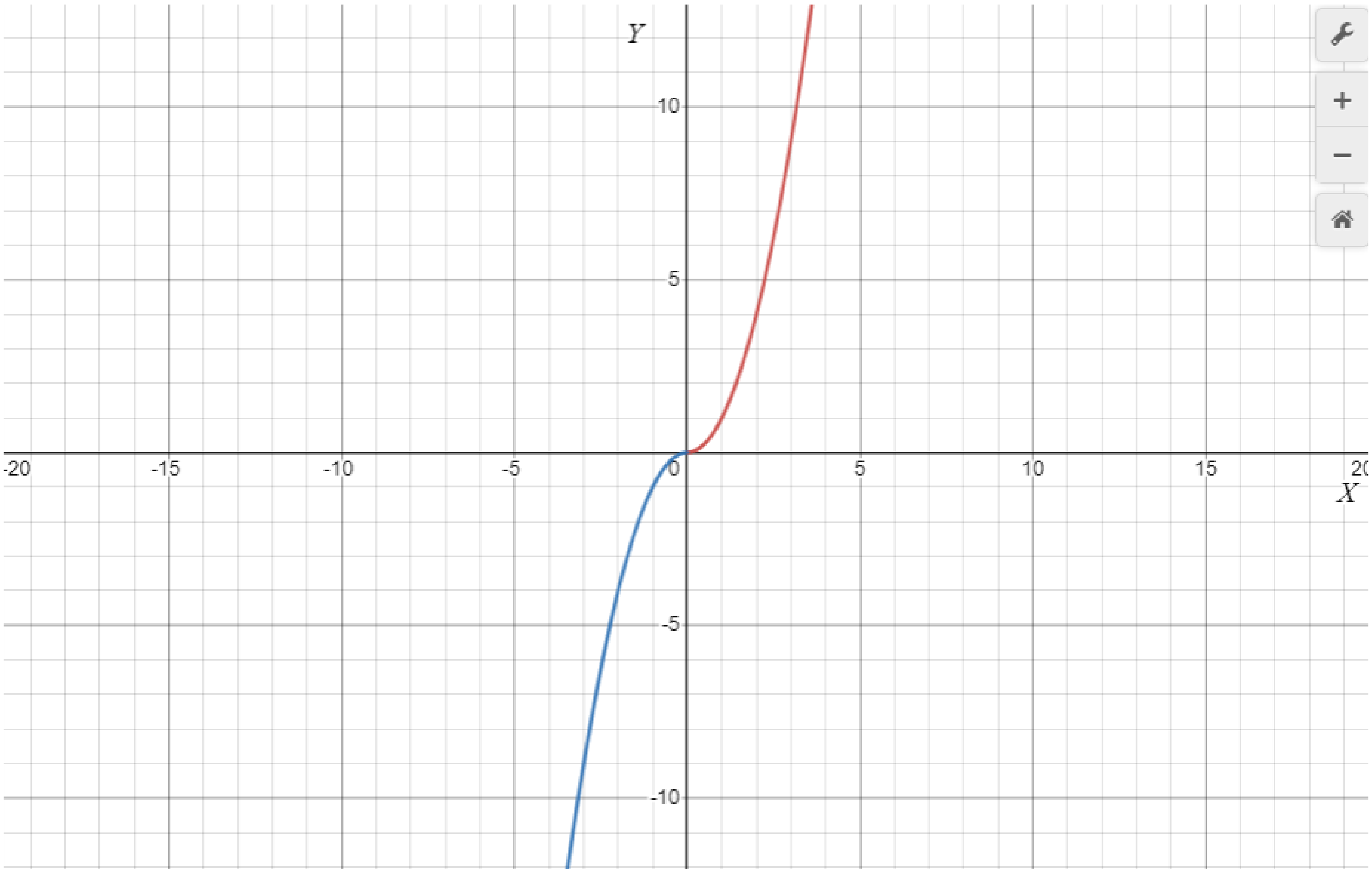
The graph of a function defined for $x \geq 0$ is given. Complete the graph for $x < 0$ make (a) an even function and (b) an odd function.



Even and Odd



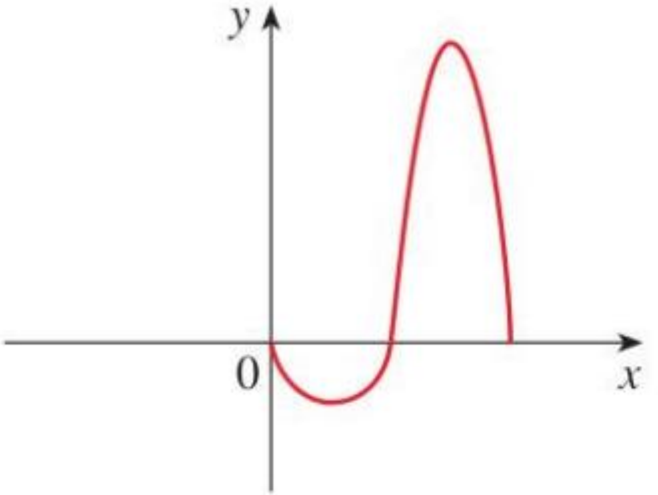
ODD FUNCTION



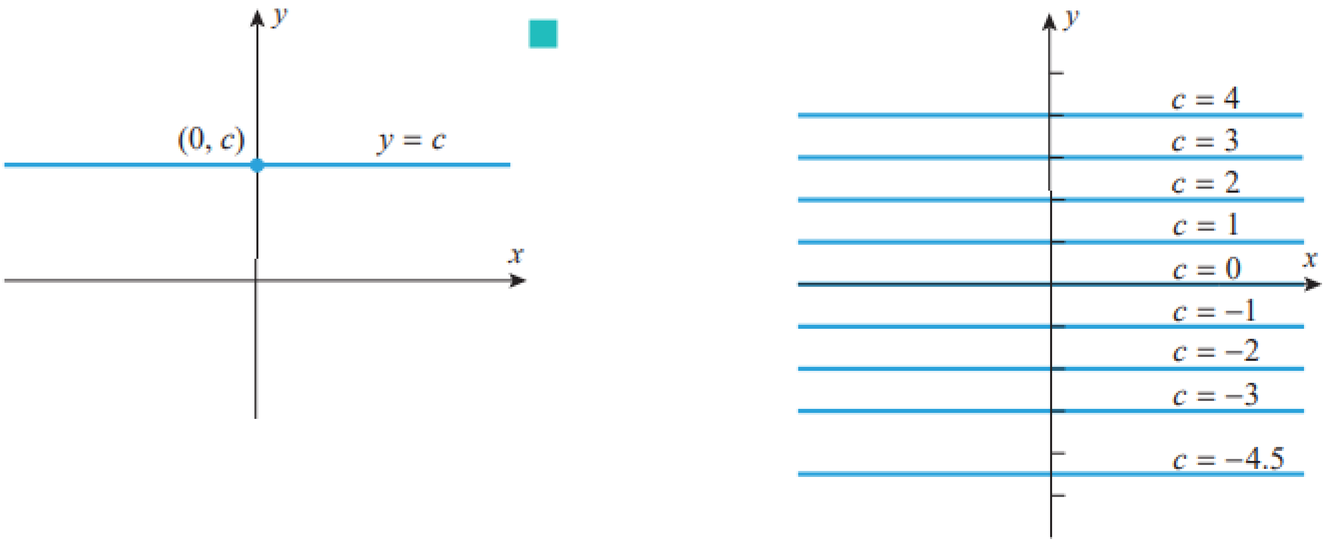
EVEN AND ODD FUNCTIONS

The graph of a function defined for $x \geq 0$ is given. Complete the graph for $x < 0$ to make (a) an even function and (b) an odd function.

Figure 3:



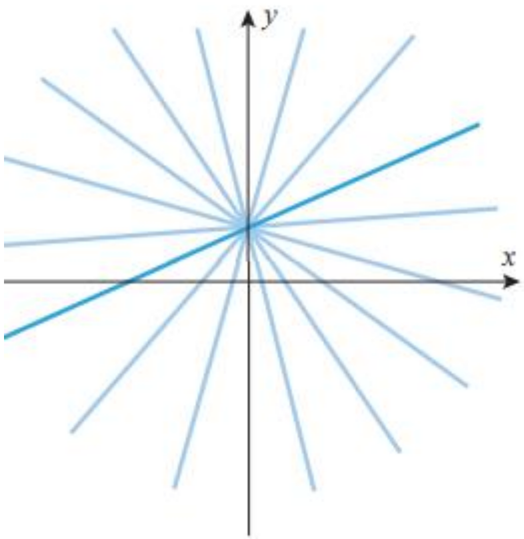
0.3 FAMILIES OF FUNCTIONS



(b)

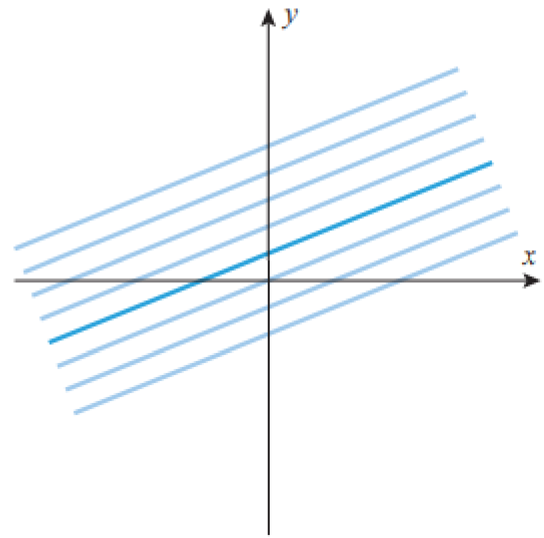
▲ Figure 0.3.1

0.3 FAMILIES OF FUNCTIONS



The family $y = mx + b$
(b fixed and m varying)

(a)



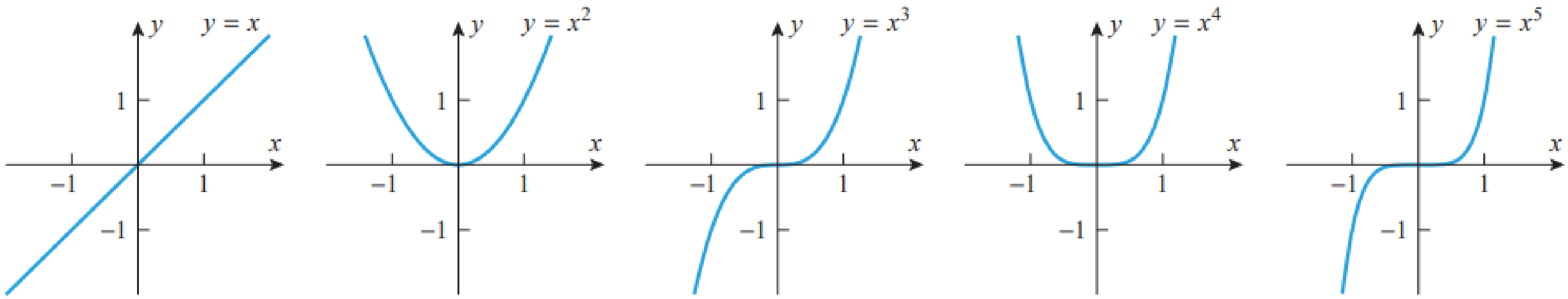
The family $y = mx + b$
(m fixed and b varying)

(b)

POWER FUNCTIONS;

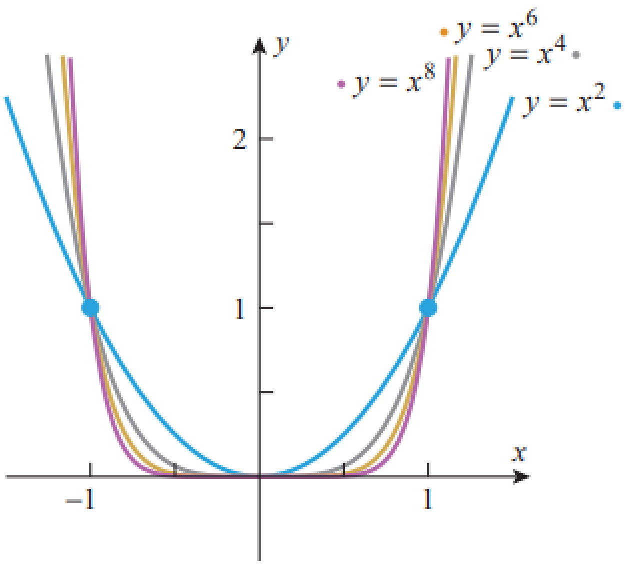
POWER FUNCTIONS; THE FAMILY $y = x^n$

A function of the form $f(x) = x^p$, where p is constant, is called a ***power function***. For the moment, let us consider the case where p is a positive integer, say $p = n$. The graphs of the curves $y = x^n$ for $n = 1, 2, 3, 4$, and 5 are shown in Figure 0.3.3. The first graph is the line with slope 1 that passes through the origin, and the second is a parabola that opens up and has its vertex at the origin (see Web Appendix H).

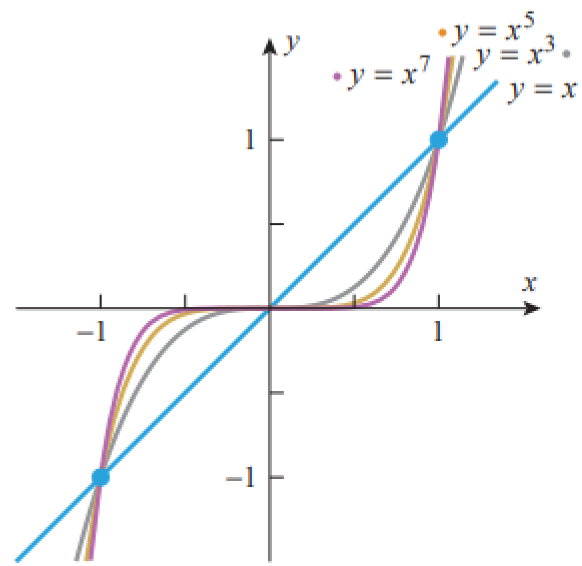


The family $y = x^n$
(n even)

For even values of n, the functions $f(x) = x^n$ are even, so their graphs are symmetric about the y-axis.



The family $y = x^n$
(n even)

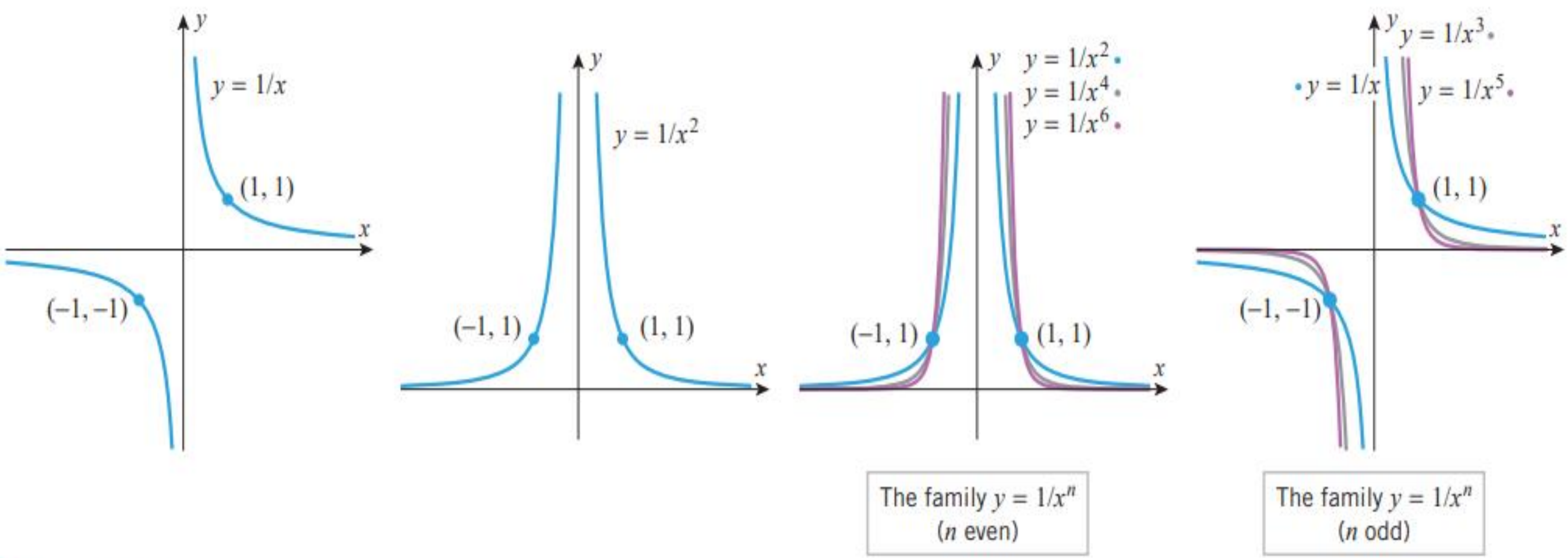


The family $y = x^n$
(n odd)

THE FAMILY $y = x^{-n}$

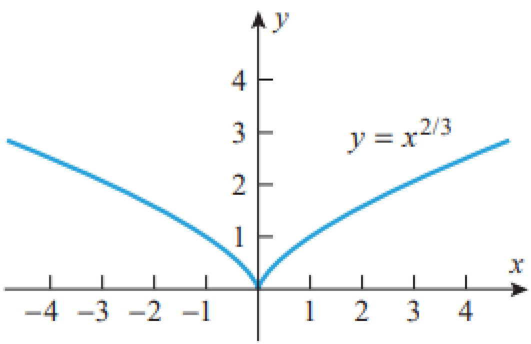
If p is a negative integer, say $p = -n$, then the power functions $f(x) = x^p$ have the form $f(x) = x^{-n} = 1/x^n$. Figure 0.3.5 shows the graphs of $y = 1/x$ and $y = 1/x^2$. T

The graph of $y = 1/x$ is called an ***equilateral hyperbola***

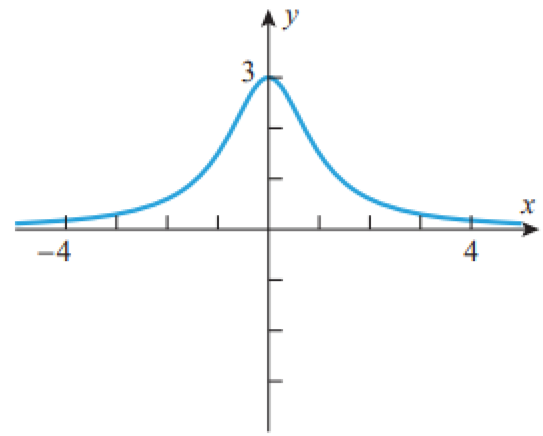


▲ Figure 0.3.5

0.3 FAMILIES OF FUNCTIONS



▲ **Figure 0.3.9**



$$y = \frac{3}{x^2 + 1}$$